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MMAE 445 Arbor Press Final Report

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Introduction

This report documents the complete design, analysis, documentation, assembly, and manufacturing workflow completed for the MMAE 445 Final Exam. The objective of this final

project was to evaluate how design intent, material selection, and manufacturing method influence the structural performance and manufacturability of an arbor press. Four design revisions were completed: Arbor Press P1 (#98471), P2 (#98472), P3 (#98473), and P4 (#98474). These revisions represent both machined and cast versions of the arbor press body, with P4 being a fully student-designed component guided by finite element analysis (FEA) results.

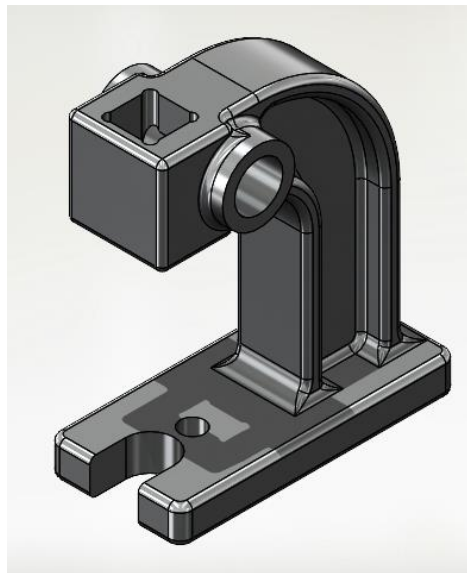
Each revision was modeled in SolidWorks, analyzed using SolidWorks Simulation under a 3 ton load case, documented with fully detailed 2D drawings, assembled into functional SolidWorks assemblies, and prepared for manufacturing using Mastercam. The final exam emphasized understanding how geometry, fillets, wall thickness, symmetry, and manufacturing constraints affect stress distribution, factor of safety, and overall part performance.

3D Modeling

The objective of the modeling portion was to create or modify 3D models for Arbor Press P2, P3, and P4 using SolidWorks. Arbor Press P1 was provided and was not modeled from scratch.

P1:

P1 is shown in the screenshot below.



Arbor Press P1 was provided as a completed SolidWorks part file and represents a machined arbor press frame. No modeling changes were made to this part. This design served as the baseline for comparison in both the FEA and design revision sections of this report.

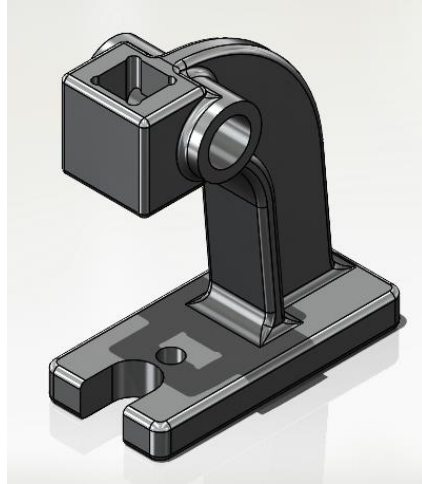
P2:

Arbor Press P2 was modeled from the provided drawing and represents a cast version of the arbor press frame. Compared to P1, P2 includes thicker wall sections at the bottom, larger fillets, and smoother transitions between features. Overall, P2 added more material in areas of high stress. These changes were made to improve stress distribution and to better suit the casting process. The screenshots below show the new changes and how the updates were made. The changes are in order like a reading a book ending with the final P2.



P3:

Arbor Press P3 is a further revised machine version of the arbor press. This part was modeled by modifying the baseline geometry to improve strength while maintaining machinability. Changes from P1 include increased fillet radii in high-stress areas and slight geometry adjustments like forward lean to better distribute load without significantly increasing weight. The screenshots below shows the new changes and the final part.



P4:

Arbor Press P4 was designed without a provided drawing. This part was created using the FEA results from P1, P2, and P3. Areas that experienced high stress in earlier designs were reinforced with thicker sections and larger fillets. The final geometry was revised until the part passed the required static study. This design represents an optimized cast arbor press frame focused on strength and manufacturability. The screenshots below show the new changes.



Discussion:

There were many differences between the four parts, most of which contributed to the strengthening of the final part with decreasing weight in some cases. The main physical differences between the parts include slanted edges for P2 and P4 as they are made to be casted. Other differences that contribute to different masses and strengths include larger fillets, material placement in more effective areas, and more. The differences in volumes can be seen below.

Mass properties of Arbor Press - P1 -Skiba Configuration: Default Coordinate system: -- default --	Mass properties of Arbor Press - P2 -Skiba Configuration: Cast-P2 Coordinate system: -- default --
Density = 0.10 pounds per cubic inch	Density = 0.26 pounds per cubic inch
Mass = 2.69 pounds	Mass = 6.56 pounds
Volume = 27.57 cubic inches	Volume = 25.22 cubic inches
Surface area = 128.41 square inches	Surface area = 114.54 square inches
Center of mass: (inches) X = 0.00 Y = 2.86 Z = 2.74	Center of mass: (inches) X = 0.00 Y = 2.47 Z = 2.58
Mass properties of Arbor Press - P3 -Skiba Configuration: Milled P3 Coordinate system: -- default --	Mass properties of Arbor Press - P4 -Skiba Configuration: Milled P3 Coordinate system: -- default --
Density = 0.10 pounds per cubic inch	Density = 0.26 pounds per cubic inch
Mass = 2.33 pounds	Mass = 7.67 pounds
Volume = 23.86 cubic inches	Volume = 29.50 cubic inches
Surface area = 113.46 square inches	Surface area = 116.12 square inches
Center of mass: (inches) X = 0.00 Y = 2.59 Z = 2.97	Center of mass: (inches) X = 0.00 Y = 2.70 Z = 2.74

FEA Studies

The objective was to conduct static stress analysis using SolidWorks Simulation.

Setup Summary:

Material: P1 and P3: 6061-T6 Aluminum (Yield Strength = 275 MPa)

Material: P2 and P4: Gray Cast Iron

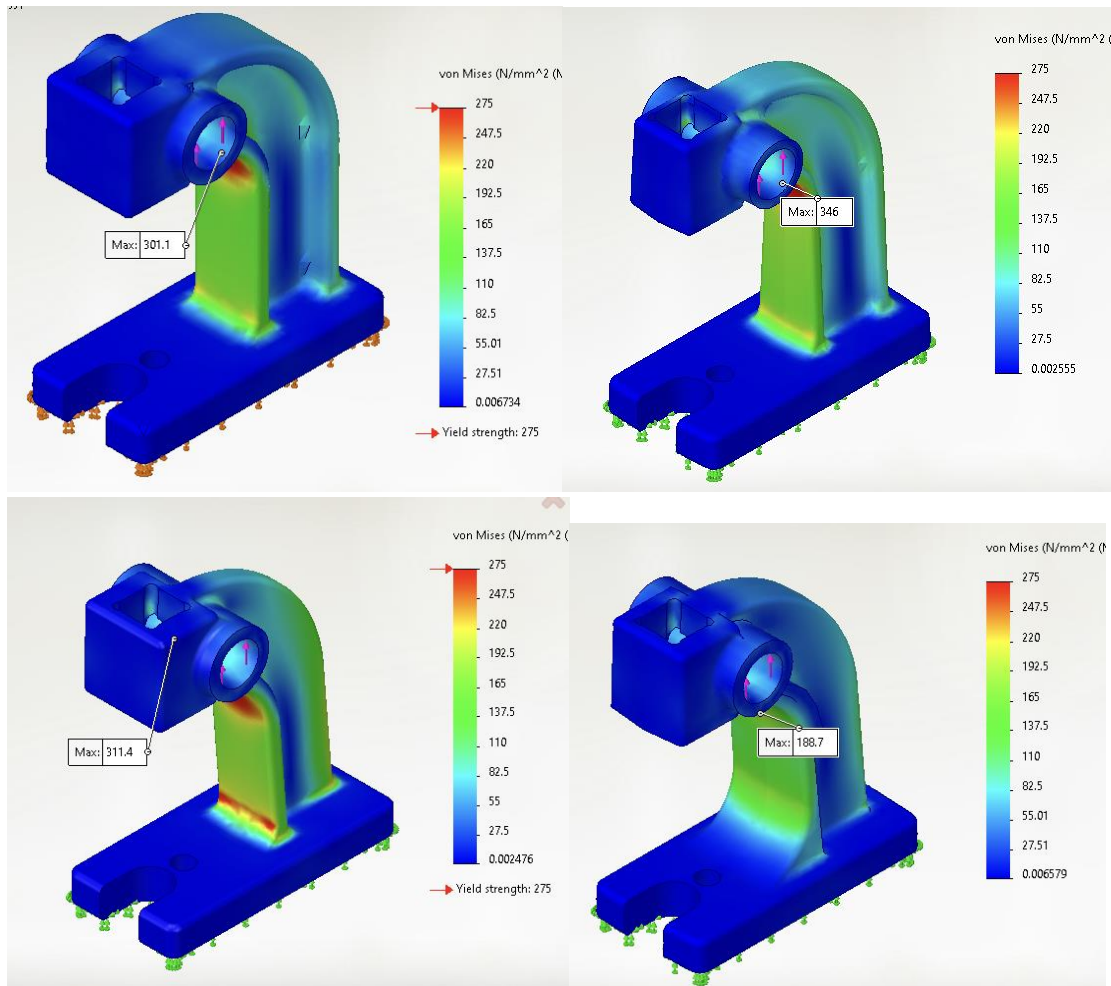
Fixtures: Entire flat bottom surface fixed

Load: 3 Ton (6,613.86 lbf)

Mesh: Default solid mesh, generated and saved as a STL.

Results:

Parameter	P1	P2	P3	P4
Max Stress (MPa)	301.1	346	311.4	188.7
Volume (in ³)	27.57	25.22	23.86	29.50

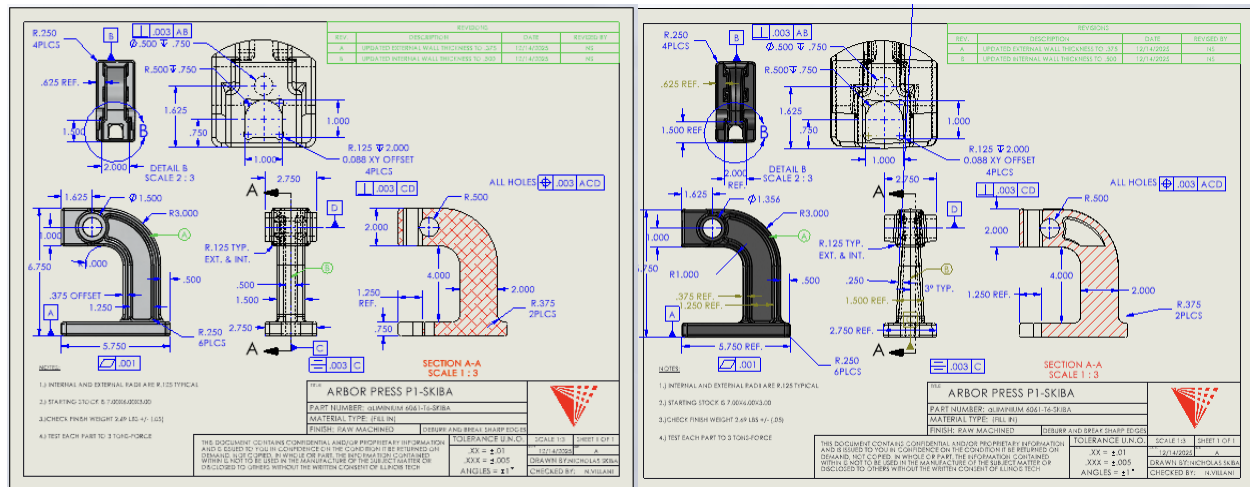


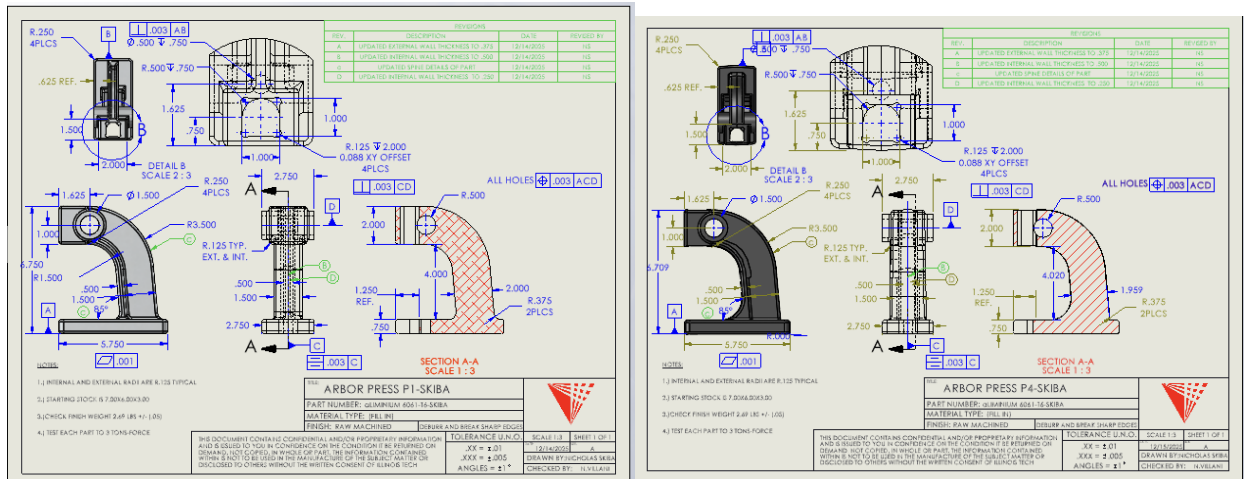
Discussion:

P1 showed high stress concentrations near sharp transitions and internal corners, resulting in a low factor of safety under the applied load. P2 showed a reduction in weight but an increase in stress, as the design changes were primarily focused on improving castability rather than optimizing structural performance. P3 improved upon P1 by redistributing material in critical areas while maintaining machinability, resulting in similar stress with a large weight decrease. P4 showed the best overall performance due to targeted reinforcement in high-stress regions identified through previous FEA results. These changes were made to improve strength, increase factor of safety, and better match the intended manufacturing method.

2D Drawings

The objective of the drawing portion was to create fully detailed SolidWorks drawings for Arbor Press P1, P2, P3, and P4 using the Student Drawing Template.



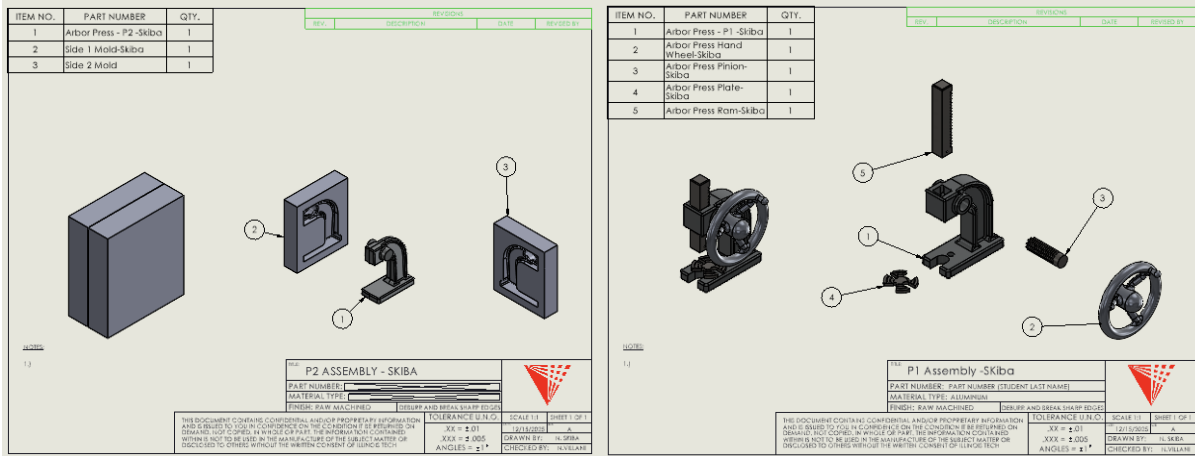


The tolerances specified by each GD&T symbol are different than the title block tolerances because the title block only provides general tolerances for non-critical dimensions. GD&T is used to control specific features that are critical to the function, strength, and assembly of the part. Features such as hole location, flat mounting surfaces, perpendicularity, and profile directly affect how the part assembles, carries load, and interfaces with other components. Because these features are more critical, they require tighter and more controlled tolerances than the default title block allows. Using GD&T ensures that these important features meet functional requirements while allowing less critical dimensions to remain within broader title block tolerances, which improves manufacturability and reduces cost.

3D Assembly

A complete 3D assembly was created for Arbor Press P1 using the provided components. The pinion and hand wheel were mated at their respective pin locations. A rack and pinion mechanical mate was applied between the pinion and the ram, with a pitch diameter of 0.834 to ensure correct gear engagement. An exploded view configuration was created and used to generate a setup sheet drawing with an isometric exploded view, bill of materials, and balloon callouts.

For Arbor Press P2, a configuration titled “Machine Details Removed” was created to suppress thru-hole features without draft. A left and right mold were then created using the cavity command. An exploded mold assembly and setup sheet drawing were also created.



Programming

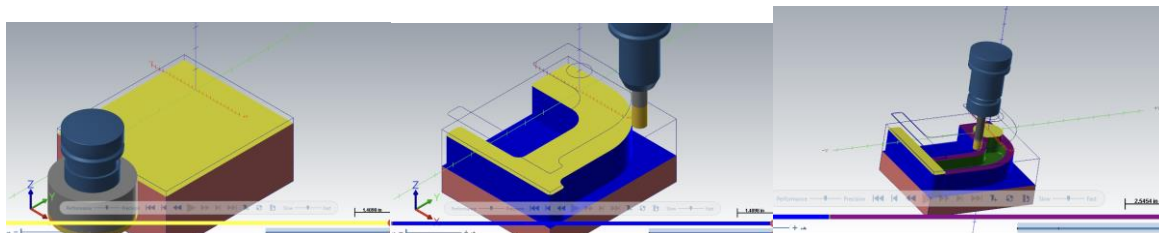
The objective of the programming portion was to generate CNC toolpaths in Mastercam for Arbor Press P1 and Arbor Press P2.

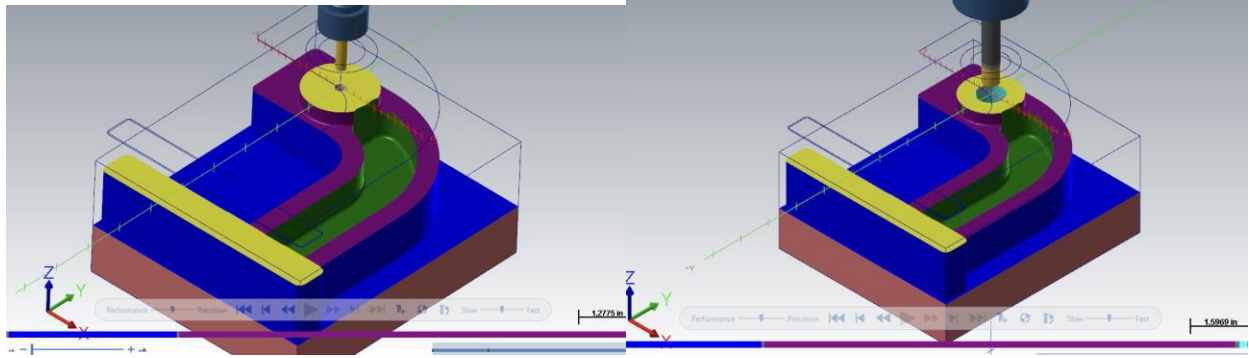
- Stock Size: 6.00 x 7.00 x 3.00 inches
- Tools: From provided tool library
- First Operation: Stock setup, roughing, finishing
- Second Operation: Stock setup, roughing, finishing
- Verification: Verified toolpaths using Color Loop simulation

Only one side of Arbor Press P1 was programmed to reflect realistic machining constraints. For Arbor Press P2, only the bottom mold was programmed.

P1

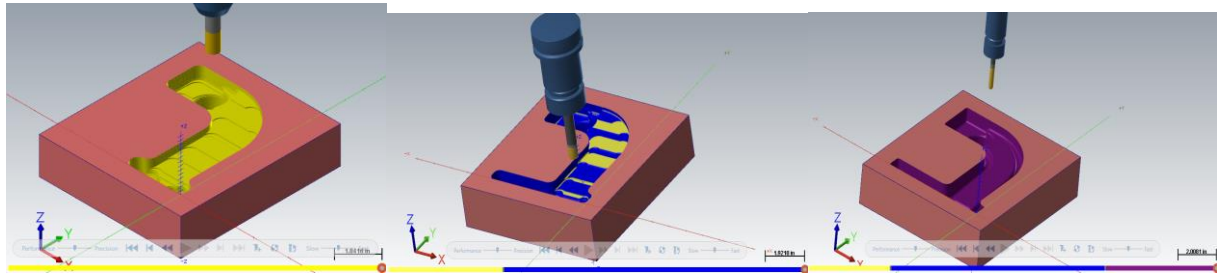
Below are screenshots of P1 first operation. This is in order from left to right going down(some toolpaths are not on here because it is finishing and looks the same). A silhouette boundary was used to get the outside edge for dynamic mill.





P2-Bottom Mold

Here is an image of P2 operations in order from left to right. The operations include dynamic Opti rough, dynamic Opti rest, and equal scallop.



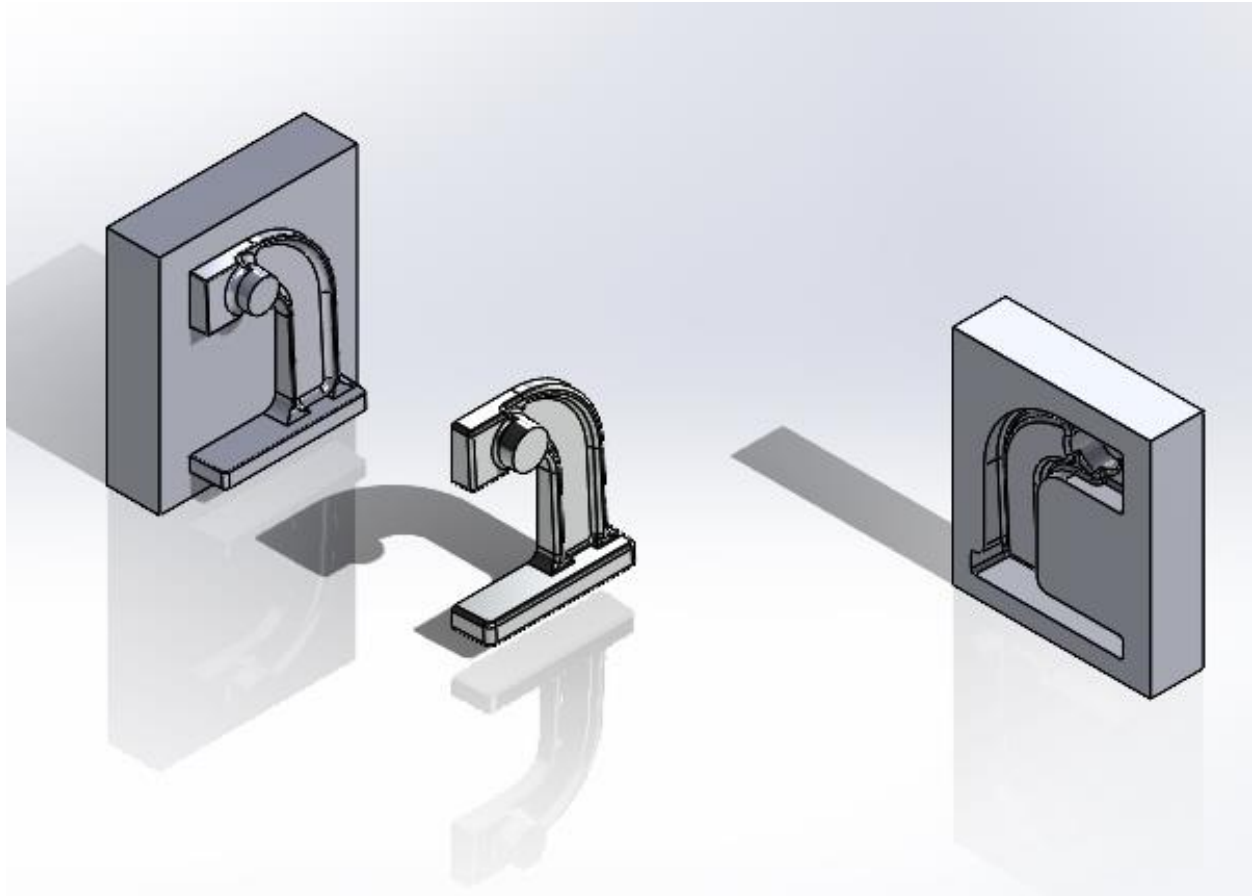
Discussion:

The thru-hole features were excluded from the bottom mold because they do not include draft and would prevent proper mold release. Including these features would cause interference during part removal and could damage the mold. Removing these features allows the part to be cast successfully and machined afterward if needed.

Extra Credit

For the extra credit portion, a core and cavity assembly was created using a shelled version of Arbor Press P2 in SolidWorks. The shelled feature was used to represent the internal geometry of the cast part while maintaining consistent wall thickness. This approach allows the core and cavity to accurately reflect the final cast geometry and ensures proper material flow and part formation during the casting process. Creating both the core and cavity demonstrates an

understanding of how internal and external features of a cast part are formed.



The core mold component, which represents the positive geometry of the cast part, was programmed in Mastercam. Toolpaths should be created using stock setup, roughing, and finishing operations to fully machine the core geometry. Verification can be performed to ensure correct material removal and surface quality. Programming the core mold highlights the differences between machining a final part and machining tooling used for casting, and reinforces the importance of precision when manufacturing mold components.

Conclusion

This final exam provided a complete workflow experience from analysis through manufacturing. Comparing P1 through P4 demonstrated how design revisions, material selection, and manufacturing methods significantly affect structural performance and manufacturability. The FEA results guided design decisions and led to improved factor of safety and stress distribution in later designs. The integration of CAD, FEA, assemblies, and CNC programming reinforced the importance of designing parts that are both structurally sound and manufacturable.